PROJECT 1 · MACHINE LEARNING

Predicting LendingClub Loan Interest Rates

Application-time prediction of int_rate — exploratory analysis, model comparison, and explainability on 100,000 loans.

100,000 LOANS · 38 FEATURES · BEST RMSE 3.87

[Course / Section] Group SAXA 8

[Team member names]

Context & Problem

Estimate the interest rate a LendingClub loan would be offered, using only information available at the moment of application.

PROBLEM STATEMENT

- Predict int_rate (loan APR, %) for 10,000 held-out applications and minimize test-set RMSE.
- Target ranges 6.46–30.99%, mean 12.95, right-skewed (skew 0.85).
- Pure regression on tabular application data — no time series, no text scoring.

ASSUMPTIONS & GUARDRAILS

- loan_status is post-origination leakage and is dropped before any modeling.
- grade / sub_grade / installment are absent from this slice — no shortcut to APR.
- No date columns exist, so validation uses a random 80/20 split, not a temporal one.

DATASET AT A GLANCE

100,000	10,000	38	6.46–31%	12.95%
Training loans	Test loans	Model features	Target range	Mean rate

Feature Engineering

One stateless pipeline turns raw application fields into 36 model-ready features, identically on train and test, keeping target information out of every fold.

01 · DATA PREP
Cast NA numerics; drop leakage

02 · FEATURE ENGINEERING
32 stateless + 4 fold-safe encodings

03 · VALIDATION
80/20 holdout + 5-fold CV

6 Credit quality & recodes
fico, fico_band, term_months

Collapses collinearity; adds geography

5 Affordability ratios
loan_to_income, bal_to_income

Scales credit to ability to pay

8 Bands & log tails
dti / util bands, log1p tails

Captures nonlinear risk; cuts skew

6 Interactions
term_x_dti, fico_x_term, util_x_fico

Credit quality × loan structure

7 Missingness & history
has_* flags, derog_score, file_age

Treats missing events as signal

4 Leakage-safe encodings
addr_state_te, purpose_te, zip3_te

Out-of-fold high-cardinality signal

Guardrail: loan_status dropped before modeling; encodings fit inside folds; raw FICO bounds, term, emp_length, title & zip replaced by engineered forms.

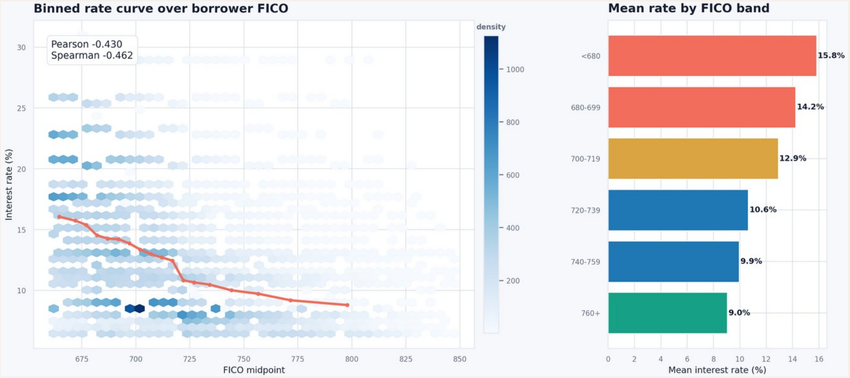
What the Data Says

FICO is the dominant application-time signal; utilization, DTI and loan term carry most of the remaining recoverable variance.

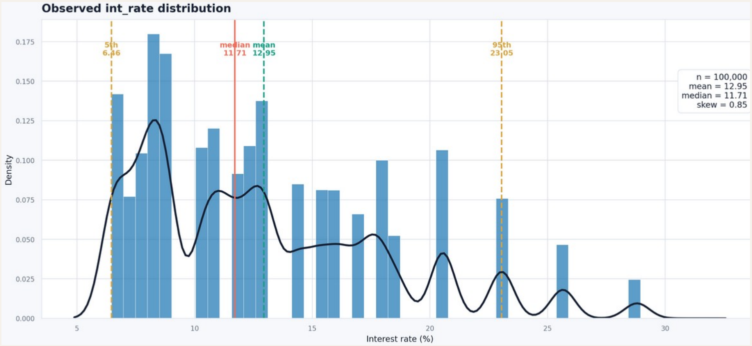
KEY FINDINGS

- FICO ↔ rate: Spearman −0.46 — higher score, lower APR.
- Utilization next: all_util +0.33, revol_util +0.30; DTI +0.18.
- 60-month term and small-business / educational purposes price higher.
- mths_since_last_record is 90.9% missing — "no record" is signal.
- Test loans are smaller (loan_amnt PSI 0.17), slightly higher FICO.

FICO vs INTEREST RATE



TARGET DISTRIBUTION



Five Models, One Winner

All five families share the same engineered features and 80/20 split. CatBoost gives the lowest validation RMSE and is the submitted model.

Model	CV RMSE	Val RMSE	Val MAE	Val R^2	Notes
Decision Tree	—	—	—	—	Single-tree baseline
Random Forest	4.072	4.019	3.007	0.410	Bagged trees; slight overfit
[5th model]	—	—	—	—	—
Neural Net (MLP)	5.101	—	—	0.06	Unstable fold; rejected
CatBoost (final)	3.863	3.872	2.902	0.465	Optuna-tuned — selected model

Tree ensembles beat the single-tree and neural-net baselines; CatBoost wins on validation RMSE (3.87) and MAE (2.90) and is used for the test-set submission.

Final Model: CatBoost

Gradient-boosted trees with native categorical handling; 5-fold per-fold target encoding gives an honest cross-validated estimate that matches the holdout.

HELD-OUT PERFORMANCE (20%)

3.863
OOF RMSE (CV)

3.872
Validation RMSE

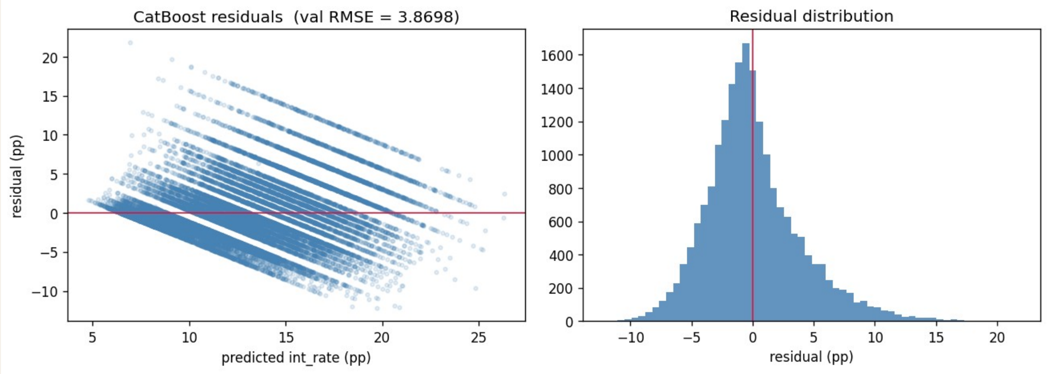
2.902
Validation MAE

0.465
Validation R^2

TUNED HYPERPARAMETERS

- learning_rate 0.03 · depth 7 · l2_leaf_reg 5.0
- random_strength 1.0 · bagging_temp 0.5 · border_count 200
- 2,500 iterations, early stopping 100; preds clipped to [6, 31].

RESIDUAL DIAGNOSTICS

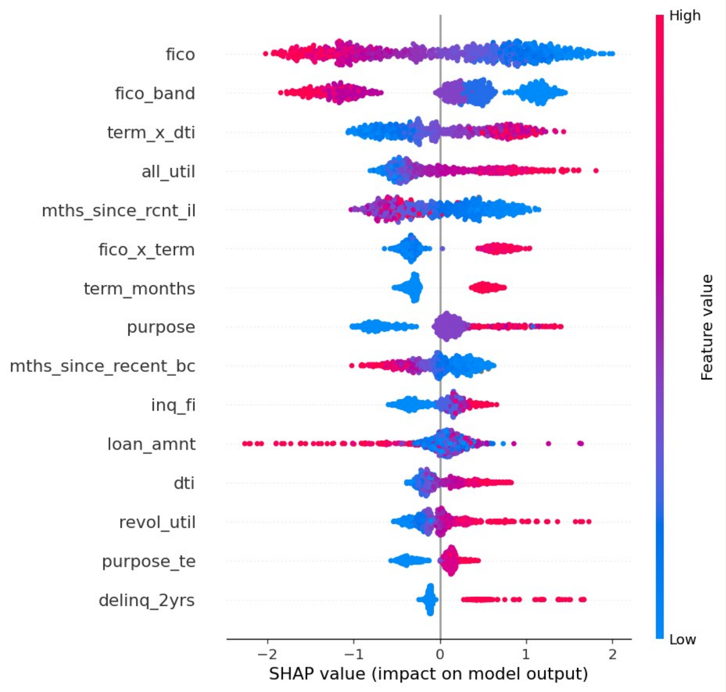


OOF RMSE (3.863) and holdout RMSE (3.872) agree closely — the model is well-calibrated and not over-fit. R^2 of 0.465 reflects the dataset's real ceiling without grade / sub-grade.

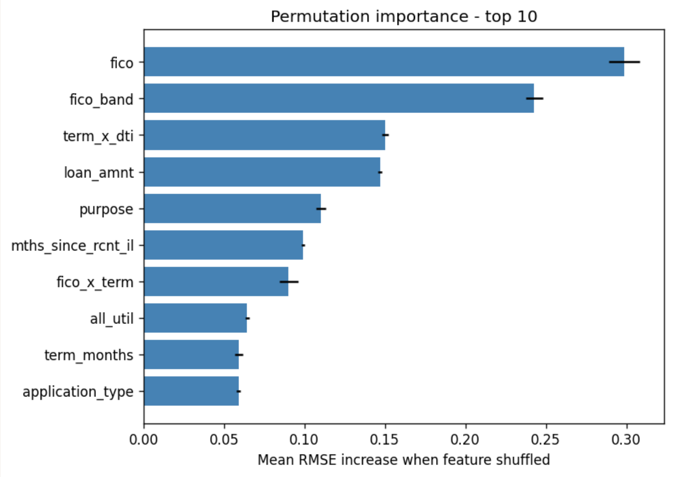
Why the Model Decides

Two techniques agree with the EDA: SHAP values and permutation importance both rank credit quality, loan structure and utilization at the top.

SHAP SUMMARY



PERMUTATION IMPORTANCE



BUSINESS INSIGHTS

- FICO dominates — a strong score is the single biggest lever on the offered rate.
- Loan structure (term × DTI, loan_amnt) is the next driver of pricing.
- Utilization & recent credit activity push rates up at the margin.
- Same drivers across SHAP, permutation & gain — the model is trustworthy.

Conclusion & Reflection

WHAT WE FOUND

- CatBoost predicts int_rate at RMSE 3.87 / R^2 0.465 on held-out data, with CV and holdout in close agreement.
- FICO, loan term and utilization explain most of the recoverable signal once leakage is removed.
- Disciplined leakage handling matters more than model complexity on this slice.

LESSONS LEARNED

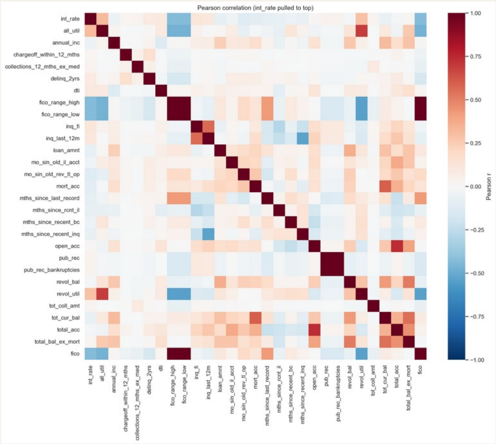
- Audit for leakage first — dropping loan_status / grade shaped the whole problem.
- Fold-internal target encoding is essential to keep the CV estimate honest.
- Know the ceiling: without grade/sub-grade, R^2 near 0.47 is the realistic limit, not a modeling failure.

Final submission: **CatBoost · test RMSE-optimized · predictions clipped to [6, 31]%**

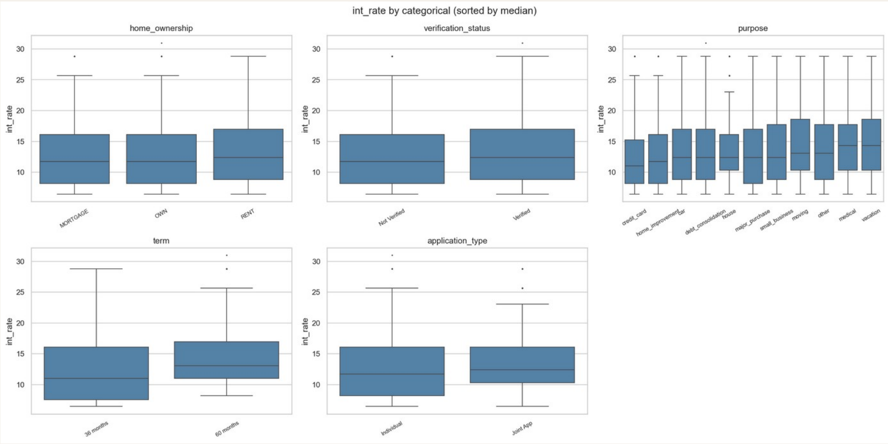
APPENDIX A

EDA Exhibits

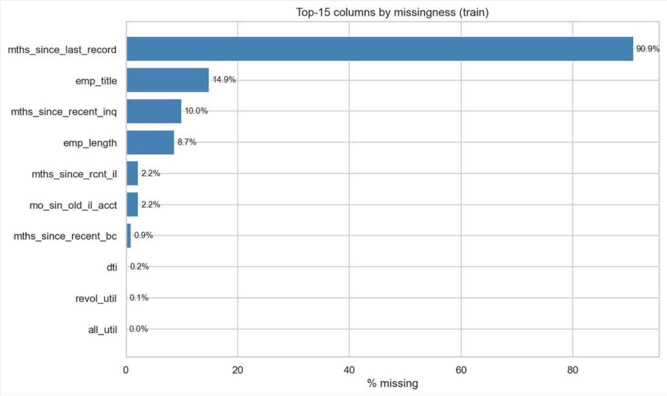
CORRELATION STRUCTURE



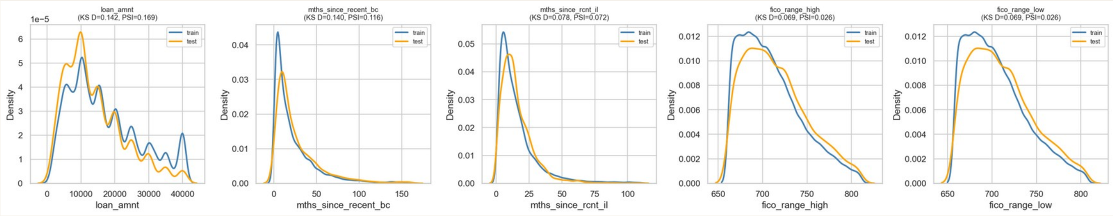
RATE BY CATEGORY



MISSINGNESS (TOP COLUMNS)



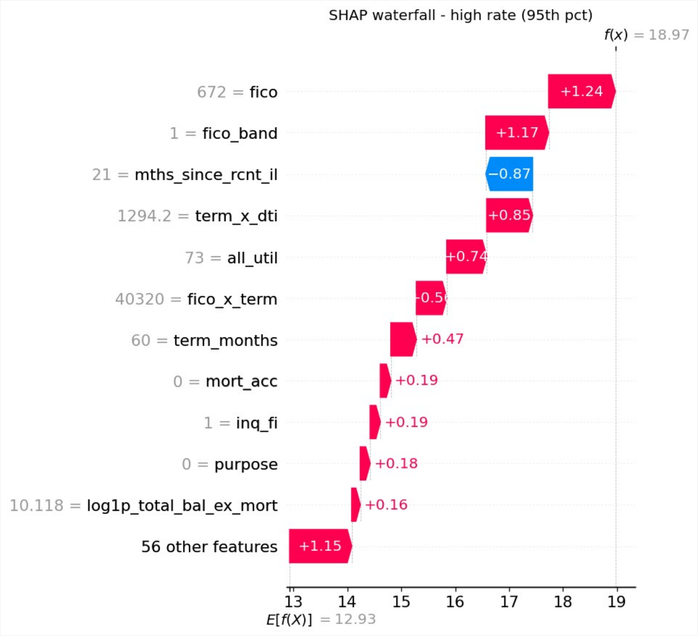
TRAIN vs TEST DRIFT



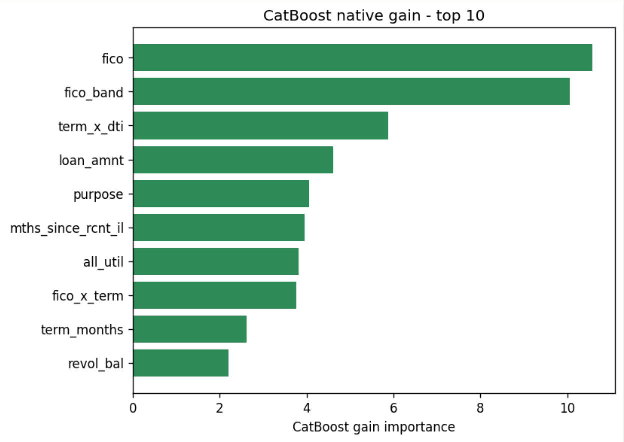
APPENDIX B

Explainability Exhibits

SHAP WATERFALL (HIGH-RATE LOAN)



CATBOOST GAIN (TOP 10)



PARTIAL DEPENDENCE (TOP 3)